Matrix Inverse via LU factorization

input : $n \times n$ matrix $[A]$
output: $[A]^{-1}$

```
[L], [U] ← lu_decompose([A])
for k ← 1 to n do
    // {e_k} is the kth column of [I]:
    // all zeros, with a 1 in row k
    A^{-1}[:, k] ← lu_solve([L], [U], {e_k})
end
```